<u>Assignment – 1.3</u>

<u>Q - 1.</u> Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

Feature	**DDR3**	**DDR4**	**DDR5**
Release Year	2007	2014	2020
Speed Range	800–2133 MT/s	2133–3200 MT/s	4800–8400+ MT/s
Voltage	1.5V (1.35V for DDR3L)	1.2V	1.1V
Power Consumption	Higher	Lower than DDR3	Lowest
Memory Capacity	Lower	Higher than DDR3	Highest
Performance	Good for older systems	Faster and more efficient	Fastest and most efficient
Typical Use Cases	Older PCs, basic computing	Modern desktops, gaming, office work	High-end gaming, content creation, AI, and future systems
Compatibility	DDR3 motherboard only	DDR4 motherboard only	DDR5 motherboard only

<u>Q - 2.</u> Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully.

Q - 3. List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

Feature	**HDD**	**SSD**	**NVMe SSD**
Speed	Slow	Fast	Very Fast
Storage Type	Spinning disks	Flash memory	Flash memory (PCIe)
Best For	Large file storage	Everyday use	Gaming, video editing, heavy workloads
Use Case / App	Store movies, photos, and backups	**Google Chrome** – Faster startup and web browsing	**Adobe Premiere Pro** – Faster video editing, rendering, and file loading

<u>Q - 4.</u> Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly.